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Stewart et al.

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(54) **STRAWBERRY PLANT NAMED 'DRISSTRAWNINETYSIX'**

(50) Latin Name: ***Fragaria* x *ananassa***
Varietal Denomination: **DrisStrawNinetySix**

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See application file for complete search history.

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(57) **ABSTRACT**

A new and distinct variety of strawberry plant named 'DrisStrawNinetySix', particularly selected for its compact plant habit, uniform fruit appearance, high yield potential, plant health, and flavor of fruit, is disclosed.

6 Drawing Sheets

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STRAWBERRY PLANT NAMED 'DRISSTRAWNINETYSIX'

Latin name:
Botanical classification: *Fragaria* x *ananassa*.
Varietal denomination: The varietal denomination of the claimed variety of strawberry plant is 'DrisStrawNinetySix'.

BACKGROUND OF THE INVENTION

Cultivated strawberry is a hybrid species of the genus *Fragaria* that is grown worldwide for its fruit. Modern strawberry was first bred in Brittany, France, in the 18^{th} century by crossing *Fragaria virginiana* with *Fragaria chiloensis*. Strawberry fruit is an aggregate accessory fruit, with the fleshy part of the fruit being derived from the receptacle that holds the ovaries.

Strawberry varieties vary widely in color, size, shape, flavor, season of ripening, degree of fertility, and susceptibility to disease. Certain varieties vary in foliage, and some vary in the relative development of their reproductive organs. Typically, strawberry flowers appear hermaphroditic in structure, but function as either male or female. Generally, commercial production of strawberry plants involves propagation from runners and distribution as either plugs or bare root plants. Cultivation is either perennial or annual plasticulture. During the off season, strawberries can also be produced in greenhouses.

Strawberry fruit is widely appreciated for its characteristic bright red color, aroma, juicy texture, and sweetness. Strawberry fruit is a popular fruit that is generally consumed either fresh or in prepared foods, such as preserves and baked goods.

Strawberry is an important and valuable fruit crop. Accordingly, there is a need for new varieties of strawberry plants. In particular, there is a need for improved varieties of strawberry plant that are stable, high yielding, and agronomically sound.

SUMMARY OF THE INVENTION

In order to meet these needs, the present invention is directed to an improved variety of strawberry plant. In particular, the invention relates to a new and distinct variety of strawberry plant (*Fragaria* x *ananassa*), which has been denominated as 'DrisStrawNinetySix'.

Strawberry plant variety 'DrisStrawNinetySix' originated from a controlled cross between the proprietary female parent '174U 4' (unpatented) and the male parent 'DrisStrawSixtySix' (U.S. Plant Pat. No. 31,703). Progeny plants from this cross, including 'DrisStrawNinetySix', were asexually propagated via stolons in Shasta County, Calif. in April 2015. Strawberry plant variety 'DrisStrawNinetySix' was later specifically identified and selected in Monterey County, Calif. in May 2016.

'DrisStrawNinetySix' was subsequently asexually propagated via stolons, and has undergone testing at test plots in Monterey County, Calif. for five years (2016 to 2021). The present variety has been found to be stable and reproduce true to type through successive asexual propagations via stolons and tissue culture.

'DrisStrawNinetySix' was particularly selected for its compact plant habit, uniform fruit appearance, high yield potential, plant health, and flavor of fruit.

DESCRIPTION OF THE DRAWINGS

This new strawberry plant is illustrated by the accompanying photographs. The colors shown are as true as can be

reasonably obtained by conventional photographic procedures. Unless otherwise indicated, the photographs are of plants that are eight months old.

FIG. **1** illustrates whole fruit of variety 'DrisStrawNinetySix'.

FIG. **2** illustrates longitudinal sections of fruit of variety 'DrisStrawNinetySix'.

FIG. **3** illustrates the upper surface of flowers of variety 'DrisStrawNinetySix'.

FIG. **4** illustrates the lower surface of flowers of variety 'DrisStrawNinetySix'.

FIG. **5** illustrates the leaves of variety 'DrisStrawNinetySix'.

FIG. **6** illustrates whole plants of variety 'DrisStrawNinetySix'.

DETAILED BOTANICAL DESCRIPTION

The following detailed descriptions set forth the distinctive characteristics of 'DrisStrawNinetySix'. The data which define these characteristics is based on observations taken in Monterey County, Calif. from 2016 to 2021. This description is in accordance with UPOV terminology. Color designations, color descriptions, and other phenotypical descriptions may deviate from the stated values and descriptions depending upon variation in environmental, seasonal, climatic, and cultural conditions. 'DrisStrawNinetySix' has not been observed under all possible environmental conditions. The botanical description of 'DrisStrawNinetySix' was taken from plants that were eight months old. The indicated values represent averages calculated from measurements of several plants. Color references are primarily to The R.H.S. Colour Chart of The Royal Horticultural Society of London (R.H.S.) (2015 edition). Descriptive terminology follows the *Plant Identification Terminology, An Illustrated Glossary*, 2^{nd} edition by James G. Harris and Melinda Woolf Harris, unless where otherwise defined.

Classification:
- *Species*.—*Fragaria* x *ananassa*.
- *Common name*.—Strawberry.
- *Denomination*.—'DrisStrawNinetySix'.

Parentage:
- *Female parent*.—Proprietary strawberry plant '174U 4' (unpatented).
- *Male parent*.—'DrisStrawSixtySix' (U.S. Plant Pat. No. 31,703).

Plant:
- *Height*.—24.3 cm.
- *Diameter*.—45.3 cm.
- *Height/width ratio*.—0.53.
- *Number of crowns per plant*.—7.4.
- *Growth habit*.—Semi-upright.
- *Density of foliage*.—Medium.
- *Vigor*.—Weak.

Stolon:
- *Number of daughter plants per mother plant*.—93.
- *Diameter at bract*.—4.0 mm.
- *Overall color*.—RHS 144B (Strong yellow green).
- *Anthocyanin coloration*.—Weak.
- *Anthocyanin color*.—RHS 39A (Strong red).
- *Density of pubescence*.—Medium.

Fruiting truss:
- *Length (from crown to base of terminal flower or fruit)*.—27.7 cm.
- *Diameter (at base of truss)*.—3.2 mm.
- *Number of berries per truss*.—4.9.
- *Attitude at first picking*.—Erect.

Color (*at base of truss*).—RHS 144B (Strong yellow green).

Leaf:

Number of leaflets.—Three only.

Color of leaf upper surface.—RHS 136A (Dark green).

Color of leaf lower surface.—RHS 138A (Moderate yellowish green).

Blistering.—Medium.

Glossiness.—Absent or weak.

Variegation.—Absent.

Terminal leaflet.—Length: 7.26 cm. Width: 7.00 cm. Length/width ratio: 1.04. Number of teeth per terminal leaflet: 18.7. Overall shape: Orbicular. Shape of base: Obtuse. Shape of apex: Rounded. Margin: Serrate to crenate. Margin profile: Flat (level with the leaflet blade). Shape in cross section: Straight.

Petiole.—Length: 188.2 mm. Diameter: 2.4 mm. Overall color: RHS 144B (Strong yellow green). Pubescence: Medium. Attitude of hairs: Upwards. Bract frequency (number present on each petiole): 0.4.

Petiolule.—Length: 8.0 mm. Diameter: 1.6 mm. Color: RHS 144B (Strong yellow green).

Stipule.—Length: 34.4 mm. Width: 14.9 mm. Stipule color: RHS 144D (Light yellow green). Anthocyanin coloration: Weak. Anthocyanin color: RHS 39A (Strong red). Pubescence: Medium.

Inflorescence:

Number of flowers per plant.—3.

Position of inflorescence in relation to foliage.—Beneath foliage.

Pedicel.—Attitude of hairs: Slightly outwards.

Flower.—Flower diameter (petal tip to petal tip on non-flattened flower): 16.8 mm. Arrangement of petals: Touching. Size of calyx in relation to corolla: Larger. Stamen: Present. Receptacle color: RHS 1A (Brilliant greenish yellow). Anther color: RHS 2A (Vivid greenish yellow).

Petal.—Length: 9.1 mm. Width: 9.3 mm. Length/width ratio: 0.98. Number of petals per flower: 5.3. Color of upper and lower surface: RHS 155D (Yellowish white). Overall shape: Orbicular. Shape of apex: Rounded. Margin: Entire. Shape of base: Concavo-convex.

Calyx.—Diameter (sepal tip to sepal tip, measured on back of flower): 25.1 mm.

Sepal.—Length: 9.3 mm. Width: 3.7 mm. Number of sepals per flower: 11.6. Overall shape: Ovate. Margin: Entire.

Flowering interval.—February to November.

Fruit:

Fruit size.—Length: 42.2 mm. Width: 35.7 mm. Length/width ratio: 1.18.

Fruit hollow.—Length: 14.3 mm. Width: 5.8 mm. Length/width ratio: 2.5.

Shape.—Conical.

Difference in shape of terminal and other fruits.—None or very slight.

Fruit color.—RHS N45B (Moderate red).

Evenness of color.—Even or very slightly uneven.

Glossiness.—Strong.

Evenness of surface.—Even or very slightly uneven.

Width of band without achenes.—Absent or very narrow.

Position of achenes.—Level with surface.

Position of calyx attachment.—Level with fruit.

Attitude of sepals.—Outwards.

Diameter of calyx in relation to diameter of fruit.—Same size.

Adherence of calyx.—Strong.

Firmness.—Firm.

Color of flesh (*excluding core*).—RHS N34B (Strong reddish orange).

Evenness of flesh color.—Uneven.

Distribution of flesh color.—Marginal and central.

Color of core.—RHS 40C (Strong reddish orange).

Sweetness/soluble solids (*in ° Brix*).—8.6.

Titratable acidity (*% as citric acid*).—0.8%.

Individual fruit weight.—23.2 g/fruit.

Achenes.—Number of achenes per fruit: 210. Weight: 0.00005 g/achene. Color of upper (sunward) side: RHS 46B (Strong red). Color of lower (shaded) side: RHS 153C (Strong greenish yellow).

Fruiting.—Harvest interval: Late March to early November. Type of bearing: Partially remontant. Productivity: 87,103 lbs to 117,135 lbs of fruit per acre per season from four- to twelve-month-old plants when grown in Aromas, Calif.

Resistance to abiotic stress, pests, and diseases:

Powdery mildew (*Podosphaera macularis*).—Moderately resistant.

Anthracnose crown rot (*Colletotrichum acutatum*).—Moderately resistant.

COMPARISON WITH PARENTAL AND REFERENCE VARIETIES

'DrisStrawNinetySix' differs from the female parent proprietary strawberry plant '174U 4' (unpatented) in that 'DrisStrawNinetySix' has greater vigor, higher yield potential, more uniform fruit shape, and better late season production than '174U 4'.

'DrisStrawNinetySix' differs from the male parent and reference variety 'DrisStrawSixtySix' (U.S. Plant Pat. No. 31,703) in that 'DrisStrawNinetySix' has weak vigor, weak stolon anthocyanin coloration, upwards attitude of petiole hairs, and medium fruit size, whereas 'DrisStrawSixtySix' has medium vigor, absent or very weak stolon anthocyanin coloration, horizontal attitude of petiole hairs, and large fruit size.

'DrisStrawNinetySix' differs from the reference variety 'DrisStrawFortyFour' (U.S. Plant Pat. No. 26,801) in that 'DrisStrawNinetySix' has weak vigor, weak stipule anthocyanin coloration, partially remontant type of bearing, and the position of the inflorescence is beneath the foliage, whereas 'DrisStrawFortyFour' has very strong vigor, strong to very strong stipule anthocyanin coloration, day neutral type of bearing, and the position of the inflorescence is above the foliage.

We claim:

1. A new and distinct variety of strawberry plant named 'DrisStrawNinetySix' as shown and described herein.

* * * * *

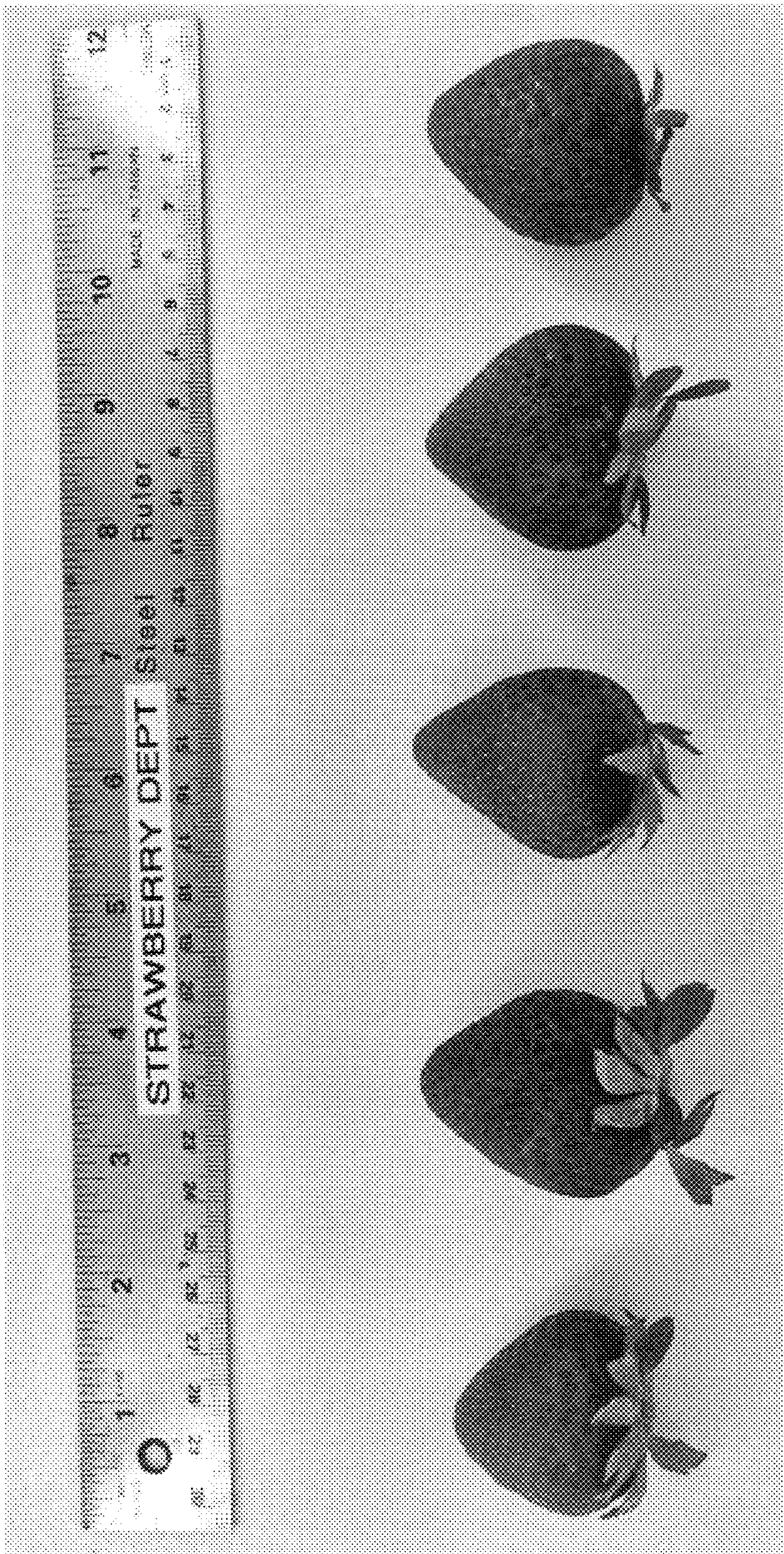

FIG. 1

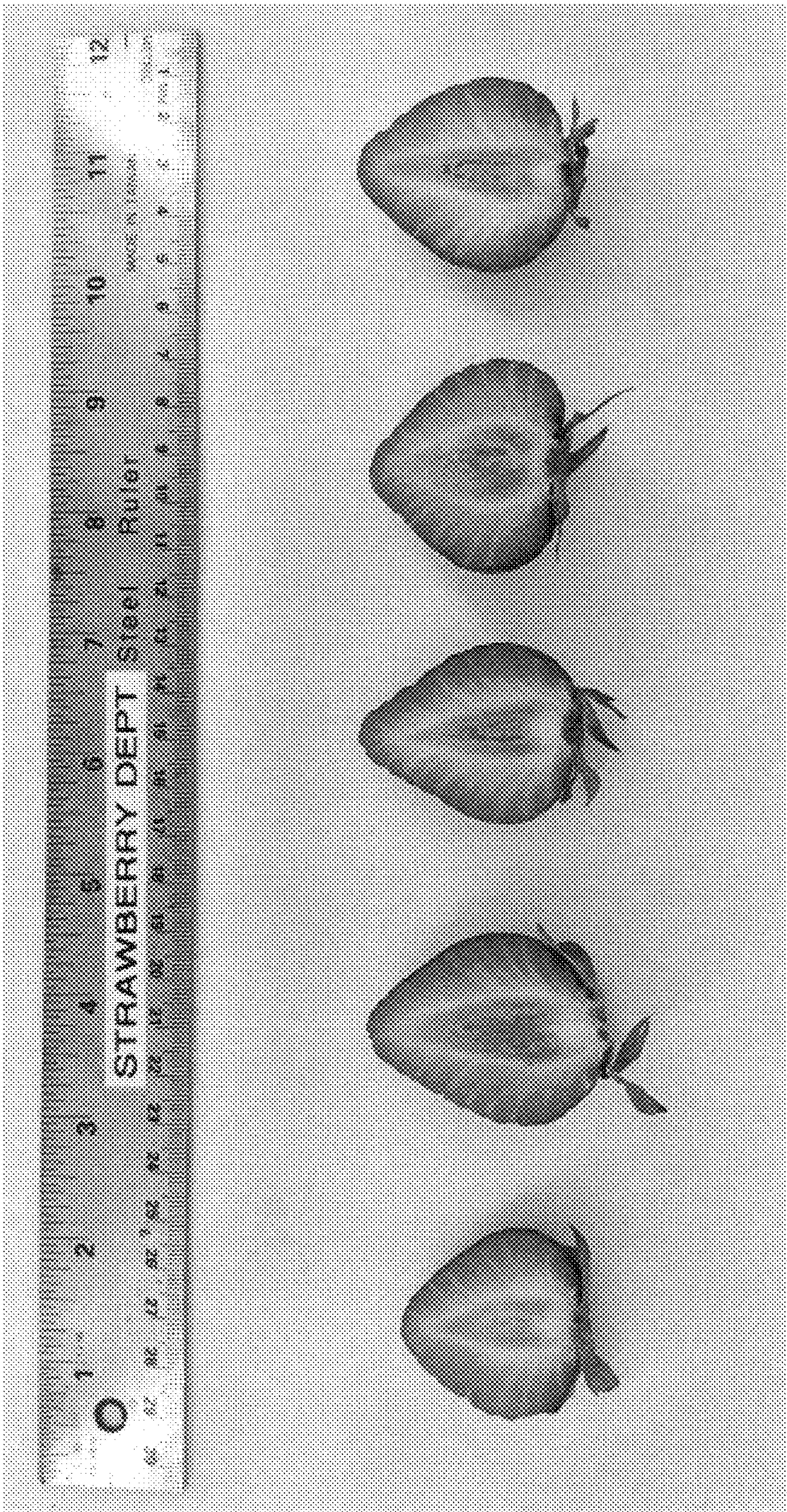

FIG. 2

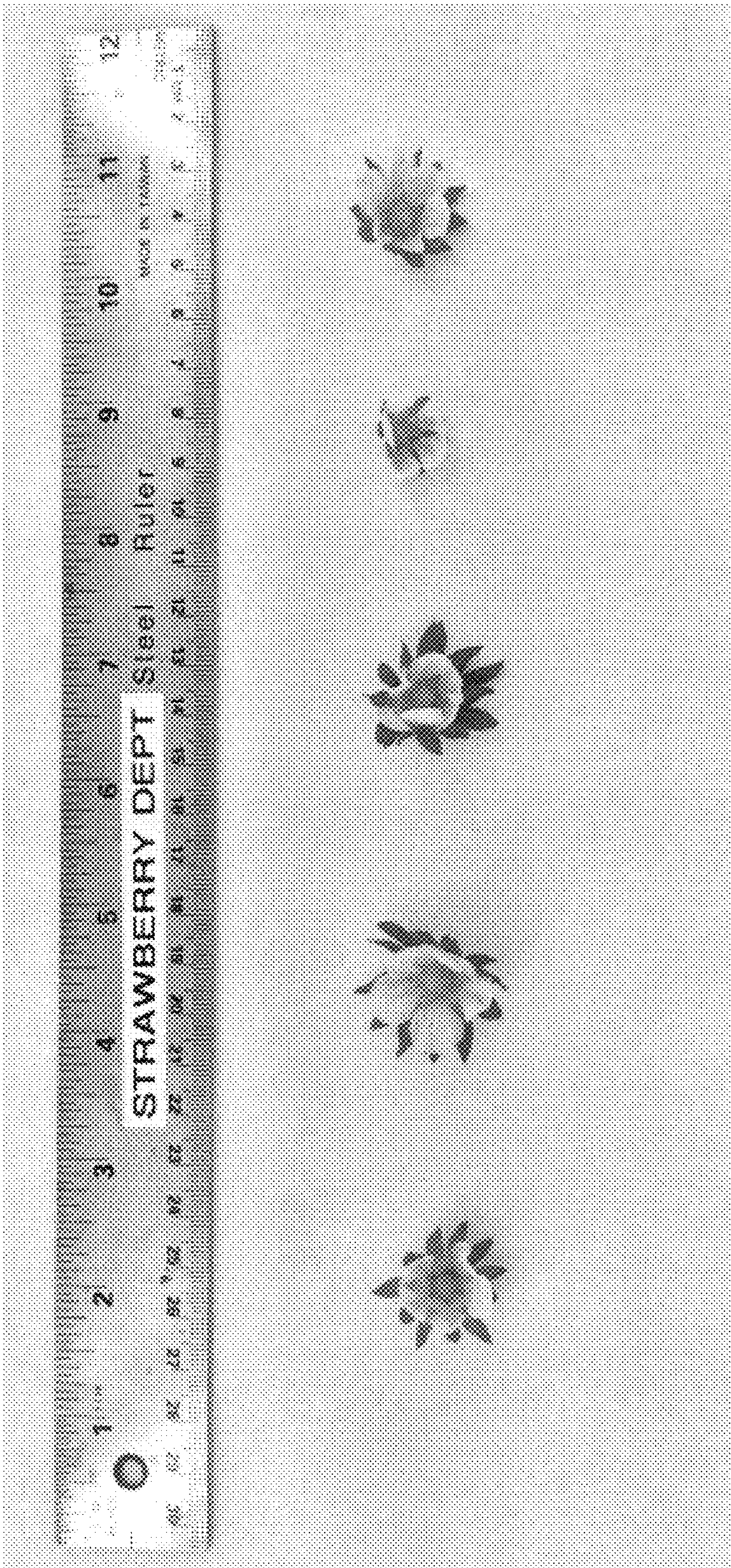

FIG. 3

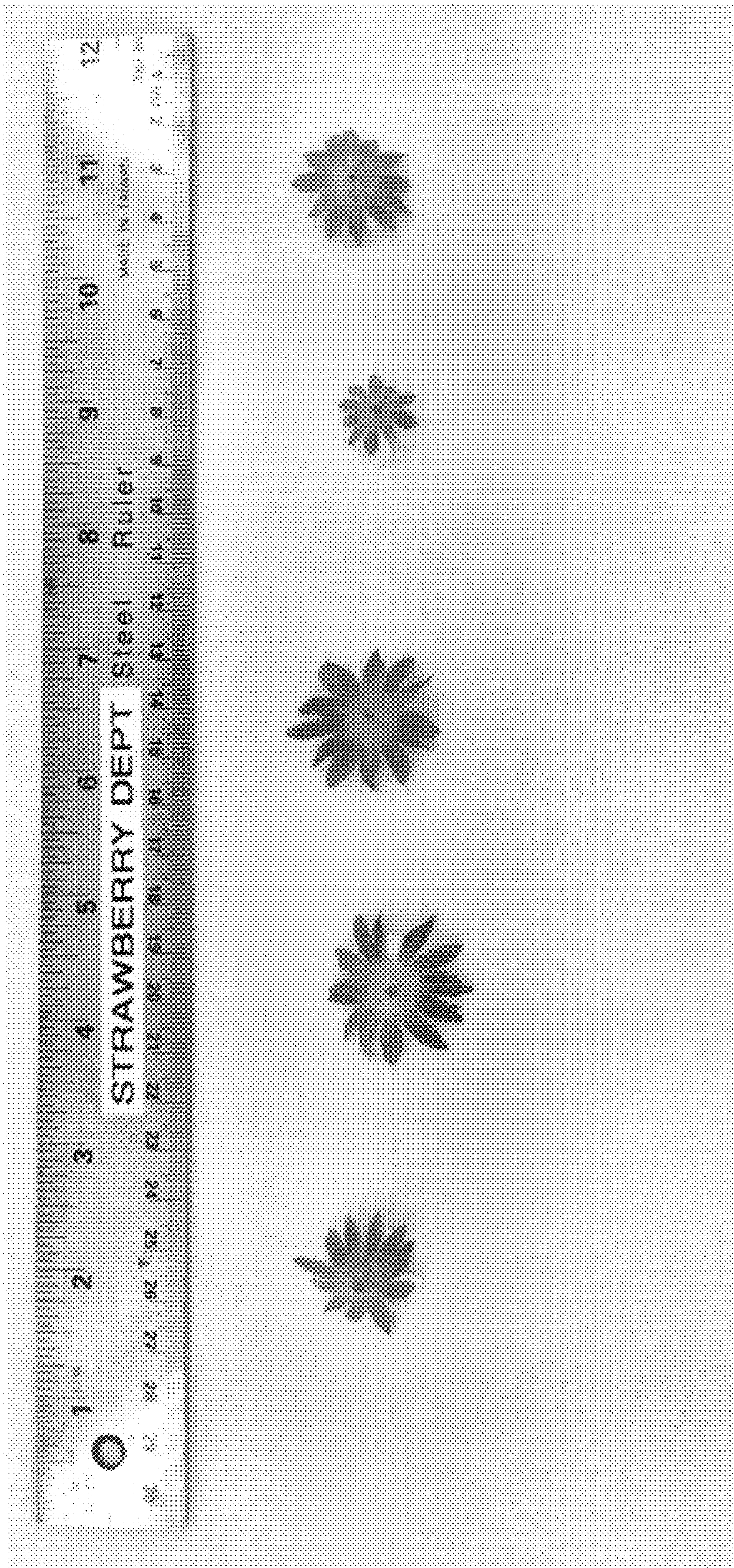

FIG. 4

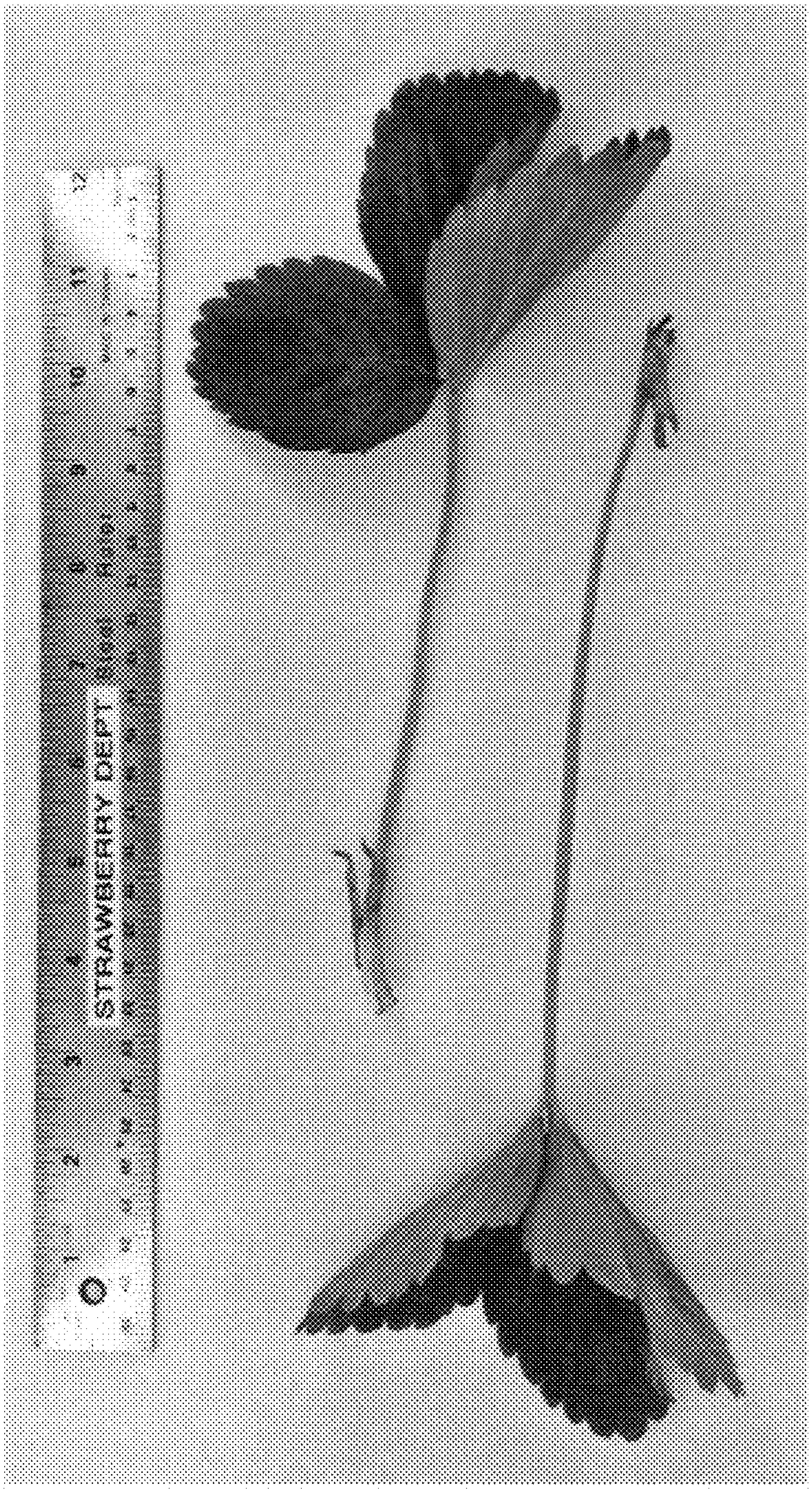

FIG. 5

FIG. 6